

2. Where does global, static, local, register variables and C Program instructions get stored?

Global , static, local : In main memory

Register variable: In registers

C program : In main memory

3. Identify the error in the below code and explain

```
#include<stdio.h>
main ()
{
    extern int i; i=20;
    printf("%d",i);
}
```

Answer: Linker Error : Undefined symbol '_i'

Explanation: extern storage class in the following declaration, extern int i; specifies to the compiler that the memory for i is allocated in some other program and that address will be given to the current program at the time of linking. But linker finds that no other variable of name i is available in any other program with memory space allocated for it. Hence a linker error has occurred .

7. What is the output of the program? Explain your answer

```
void myshow();

int main()
{
    myshow();
    myshow();
    myshow();
}

void myshow()
```

```
{  
    static int k = 20;  
    printf("%d ", k);  
    k++;  
}
```

Ans - 20 21 22

Variable of type static holds its value until the end of program execution .Static variables do not initialize again and again with function calls